

G Harbour Keys

A synthetic piano in the spirit of the early-90s ROM pianos, with a bell layer, tape wear, a mod rack and an 80s space.



Overview

G Harbour Keys is not a recreation of a real piano. It is a simulacrum of the piano sounds in the early-90s sample-based synths. Underneath is a physical model - hammers, strings, a soundboard and a bridge that lets the strings ring in sympathy - so it still answers your touch like an instrument.

On top of that model sit the things that made those ROM pianos sound the way they did: a glassy FM bell layer on every key, sample-rate and bit-depth reduction, a worn tape deck with wow and hiss, a mod rack (chorus, tremolo / auto-pan, vibe and ping-pong delay) and a room, plate or gated reverb.

Up to 16 keys can sound at once, across the 88 keys from A0 to C8. Twelve patches cover the classic territory, from bright house-style pianos to foggy, vaporwave tape keys.

Reach for it for

- Glassy late-80s and early-90s pop and house piano parts
- Vaporwave and lo-fi keys with tape wobble and a worn top end (WOW, VHS)
- Electric-piano-like tines with tremolo pan (UPRIGHT, MALLET, BELL)
- Gated-reverb ballads and big plate-drenched chords
- Crunchy, aliasing sampler textures (ROM)

Playing it

G Harbour Keys is an instrument: it has no audio input and makes its sound from MIDI.

1. Insert G Harbour Keys on a track (as the first plug-in, or after a MIDI effect).
2. Arm the track for recording and set its input to your MIDI keyboard (or draw notes in a MIDI item).
3. Play. The MIDI light at the top right flashes on each note; the PEDAL light shows when the sustain pedal is down.

It listens on all MIDI channels and plays notes from A0 to C8. It responds to the sustain pedal (CC64), the soft pedal (CC67, a softer, darker strike) and all-notes-off (CC120 and CC123). A note with velocity 1 - 3 lifts the key's dampers without striking it, so it can ring in sympathy with the other strings. The top keys (from F6 up) have no dampers, as on a real piano. Incoming MIDI is passed on down the chain.

Controls

The header

Patch display	Shows the patch number and name. A * after the number means you have changed a control since loading it. The second line shows GRAND or UPRT, the striker and PED:ON while the sustain pedal is down; while you turn a knob it shows that knob's name and value instead. Click the display for the patch list.
PATCH arrows	The upper key steps to the next patch, the lower key to the previous one. A patch sets every sound control; DRIFT,

	VOLUME, the stereo controls and the output stage are left as they are.
GRAND / UPRIGHT	The core model. GRAND has a deeper, longer-ringing soundboard and long sustain. UPRIGHT has shorter, stiffer bass strings, a smaller, higher soundboard, shorter sustain and a little more action knock. The panel changes colour with the model.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.
MIDI / PEDAL	MIDI flashes on each note; PEDAL lights while the sustain pedal is held.

Engine

FELT / TACK / PLUCK / MALLET	What strikes the strings. FELT is a normal piano hammer. TACK puts metal on the wire: a shorter, brighter strike with a metallic click - the old tack-piano trick. PLUCK catches and releases the string like a plectrum, for a harpsichord-like edge. MALLET is heavy and soft, for a round, tine-like tone.
HARDNESS	The felt, 0 - 100%: soft and dark at the bottom, compacted and bright at the top. Harder felt gives a shorter, brighter attack.
DYNAMICS	How much your velocity changes the sound, 0 - 100%. Low settings even out your playing; high settings give a wide range from soft to loud.
SUSTAIN	How long notes ring while held, 0 - 100%. It also sets how long the bell layer rings. It reaches notes that are already sounding.
UNISON	Detunes the two or three strings of each key against each other, up to 28 cents at full - honky-tonk at the top of the

	range. The display shows the detune in cents while you turn it.
BELL	The level of the FM bell layer on every key: a fast fine transient over a sine body - the glassy top of the classic house piano and the FM electric piano.
GLASS	The brightness of the bell layer (its modulation index): low is a soft sine, high a glassy, metallic chime.

ROM & Tape

ROM	Sample-rate and bit-depth reduction, 0 - 100%: from 48 kHz / 16 bit down to about 10 kHz / 8 bit, with the aliasing left in. The sound of a cheap sample ROM.
WOW	Tape speed wander, 0 - 100%: a slow wow, a faster flutter and a little random drag. The vaporwave detune.
VHS	Worn tape, 0 - 100%: the top rolls off, hiss comes up and soft tape saturation squeezes the sound.

Mod rack

CHORUS: RATE / DEPTH / MIX	A stereo bucket-brigade-style chorus. RATE 0.05 - 8 Hz, DEPTH 0 - 100%, MIX 0 - 100% (0 = off).
TREM / PAN: RATE / DEPTH	Stereo auto-pan: the left and right sides dip in turn, the classic electric-piano trick. RATE 0.1 - 12 Hz, DEPTH 0 - 100% (0 = off).
VIBE: RATE / DEPTH	A four-stage photocell phaser, its lamp sweeping the notches. RATE 0.1 - 10 Hz, DEPTH 0 - 100% (0 = off).
DELAY: TIME / FEEDBACK / MIX	A ping-pong delay: the echoes start on the left and bounce across. The feedback is warm and saturating, losing top and bottom on each repeat. TIME 20 - 1500 ms, FEEDBACK 0 - 95%, MIX 0 - 100% (0 = off).
SYNC	Click to step through MS (TIME in milliseconds), 1/4, 1/8 D (dotted), 1/8 and 1/4 T (triplet). The note values follow the

project tempo, up to 1.5 s; TIME is not used while synced.

Space

ROOM / PLATE / GATED	The reverb type. ROOM: a room of about 0.5 - 1.3 s (longer as SPACE goes up). PLATE: bright and long, about 1.4 - 4 s. GATED: the 80s gated reverb - a long, bright tail cut short about 0.3 s after each note.
SPACE	The amount of reverb, 0 - 100%. For ROOM and PLATE it also lengthens the tail.
SPREAD	Stereo spread across the keyboard, as heard from the player's seat: bass to the left, treble to the right. 0% is mono.
tone	Overall brightness, 0 - 100 (50 = neutral): a tilt around 900 Hz, and the strings themselves lose their top more slowly as you turn it up.

Drift and level

DRIFT	Gives every note its own small random tuning, up to 6 cents either way at 100% (15% by default), as on an instrument that has drifted slightly out of tune.
VOLUME	The output level, -24 to +6 dB.
SAT	Band saturation on the way out, 0 - 100%: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.
OFF / CLIP / LIMIT	The final stage: nothing, a soft clipper at -0.3 dBFS (no latency), or a lookahead limiter (3 ms latency, compensated by REAPER). CLIP is on by default.
MONO	Makes everything below this frequency mono (12 dB/oct), 20 - 500 Hz; fully left = OFF. Keeps the bass centred.
WIDTH	Raises the side signal above MONO, up to twice as wide at +100%.

In REAPER's parameter list only

The finer physical controls of the piano model have no knob on the panel, but they are in REAPER's parameter list (and every patch sets them).

Stiffness	How stiff the strings are: more stretches the upper partials sharper, for a harder, more metallic tone.
Damper	How quickly a note dies away when you let go of its key: higher settings let it ring on longer.
Body	How much of the soundboard's resonance you hear.
Sympathetic resonance	How strongly the strings share energy through the bridge: the other strings ring in sympathy, especially with the sustain pedal down.
Contact	How long the striker stays on the string: longer contact is softer and darker.
Bounce	Low settings let the action catch the hammer after the first blow; high settings let it bounce back for a second strike.
Wire age	Older strings: a duller top, shorter sustain and a little unevenness between the strings.
Tuning variance	Each key's own small tuning error (up to 6 cents), and a little spread between its strings.
Board size / Board density	Size moves the soundboard's resonances down (bigger) or up (smaller); a denser board rings longer and carries more of its upper resonances.
Lid	Open, Half or Closed: closing the lid makes the sound darker and boxier.
Mic distance	Close to far: further away is darker, quieter and narrower, with more room.
Bias / Sat low band / Sat high band / Ceiling	The rest of the output stage: the saturation's even-harmonic bias, where its low and high bands start (40 - 400 Hz and 2 - 16 kHz) and the clip ceiling or limiter threshold.

Presets

The twelve factory patches. Choose them from the patch display or step through them with the PATCH arrows.

Harbour Lights	Bright grand with tack strikers, a bell layer, a touch of ROM, chorus and a dotted-eighth delay on the plate.
Ocean Drive EP	Upright with mallets and a strong bell: an electric-piano tone with tremolo pan, chorus and a quarter-note delay.
Neon Rain	Felt grand through ROM, tape wow and VHS wear, with deep chorus and a dotted-eighth delay.
Sunset Strip	Hard, bright tack piano with a bell layer, a little ROM grit, chorus and an eighth-note delay in a small room.
Mall Fountain	Soft, long-sustaining grand through heavy wow and VHS, with vibe, chorus, a 500 ms delay and lots of plate.
Night Bus	Dark upright with mallets and the lid closed, worn tape, slow tremolo pan and a room.
Palm Glass	Grand with tacks and a loud, very glassy bell layer, chorus and delay on the plate.
Cassette Ballad	Felt grand, half lid, long sustain, gentle tape wow and VHS, chorus and plate.
Gated Romance	Bright tack grand into the 80s gated reverb, with bell and chorus.
Pastel Tine	Upright with mallets and a bright bell: a tine-like tone through vibe, light chorus and tape.
Floppy Disk	Upright tack piano crushed by heavy ROM reduction, in a small room.
Harbour Fog	Very soft, long-sustaining grand with mallets, half lid, tape wear, chorus, vibe and a long 600 ms delay in a big plate.

Tips

- Start from a patch and change the striker: the same patch with FELT, TACK or MALLET gives three quite different instruments.
- For the classic house-piano top, raise BELL and GLASS together; for a softer electric-piano feel, use UPRIGHT, MALLET and some TREM / PAN.
- ROM, WOW and VHS add up quickly: a little of each often sounds more convincing than a lot of one.
- Hold the sustain pedal and play: with Sympathetic resonance up, the undamped strings ring along with what you play.
- If the low end gets wide and woolly in a mix, set MONO to about 120 - 150 Hz.

File: gfx-g-harbour-keys.jsfx